CSC_51053_EP, Exercise Session 2: Modeling, Relational Algebra and Calculus

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1 Internship management system

We consider a system for the online management and selection of internship by students.

Each internship has an ID, a title, a category, and a description. Each internship is proposed by a supervisor; supervisors have an ID, a name, and an email. Each student has an ID, a name, and an email.

To be visible to the students, an internship must be validated by a professor responsible of the internship programme. Each professor has an ID, a name, and an email. There may be several professors jointly in charge of the internship program; it is sufficient if one of them validates the internship. The system needs to record who validated an internship, when, and an optional comment.

A student may become selected for an internship, which models that the supervisor has agreed to work with this student on the internship. A student can be selected for at most one internship, and for each internship, at most one student can be selected.

An internship is available if it is validated and no student has been selected for the internship.

1. Propose an E-R model for this information system.
2. If we want to register a student in the internship program, what does this correspond to in the E-R model?
3. In the E-R model, can we model a validation by a professor not yet registered in the system?
4. Propose a corresponding relational model.

2 Flight information [1]

Consider the following relations with airline flight information:

Flights(*flno*: integer, from: string, to: string, distance: integer, departs: time, arrives: time)
Aircraft(*aid*: integer, aname: string, cruisingrange: integer)
Certified(*eid*: integer, *aid*: integer)
Employees(*eid*: integer, ename: string, salary: integer)

Employees contain pilots but also other kinds of employees. In Certified, *eid* is a foreign key to Employees and *aid* a foreign key to Aircraft. Certified(e, a) means that employee e is certified to fly (pilot) aircraft a. Employees are pilots if and only if they are certified on at least one aircraft; a pilot may be certified for several aircrafts.

Write the following queries in relational algebra and tuple relational calculus:

1. Find the *eid*s of pilots certified for some Boeing aircraft.
2. Find the names of pilots certified for some Boeing aircraft.
3. Find the *aid*s of all aircraft that can be used on non-stop flights from Boston to Madras.
4. Identify all the flights that can be piloted by every pilot whose salary is more than $100,000.
5. Find the names of pilots who can operate planes with a range greater than 3,000 miles but are not certified on any Boeing aircraft.

6. Find the *eid*s of employees who make the highest salary.

7. Find the *eid*s of employees who make the second highest salary.

8. Find the *eid*s of employees who are certified for exactly three aircraft.

3 Cardinalities of relational expressions [1]

We denote by $|R|$ the number of tuples in relation R. Given two relations R_1, R_2, give the minimum and maximum number of tuples which may belong to the results of the following relational algebra expressions; also, specify all constraints which should apply for each expression to be correct. (1) $R_1 \cup R_2$; (2) $R_1 \cap R_2$; (3) $R_1 \setminus R_2$; (4) $|R_1| \times |R_2|$; (5) $\sigma_{a=5}(R_1)$; (6) $\pi_a(R_1)$; (7) R_1/R_2.

4 Expressivity

1. Show how to express intersection ($\cap$) using only operators among: select (σ), project (π), union ($\cup$), cartesian product ($\times$) and set difference ($\setminus$).

2. Show how to express relational division ($/$) using only operators among: select (σ), project (π), union ($\cup$), cartesian product ($\times$), and set difference ($\setminus$).

3. Prove that any relational algebra expression can be equivalently expressed in tuple relational calculus.

4. A query Q is *monotone* iff for any database D_1 and D_2 such that D_1 is included in D_2 then $Q(D_1)$ is included in $Q(D_2)$. Prove that for any query expressed using with the operators $\pi, \sigma, \bowtie, \cup$ and $\times$, this query is monotone.

5. Prove that the set of queries expressed by any relational algebra expression is not equivalent to the set of queries expressed using with the operators $\pi, \sigma, \bowtie, \cup$ and $\times$.

References

[1] Raghu Ramakrishnan and Johannes Gehrke. *Database management systems (3rd ed.)*. McGraw-Hill, 2003.